

Green concrete partially comprised of rice husk ash as a supplementary cementitious material – A comprehensive review

Blessen Skariah Thomas*

Department of Civil Engineering, Providence College of Engineering, Chengannur, Kerala, India

ARTICLE INFO

Keywords:

Waste management
Global warming
Rice husk ash
Cement concrete
Durability

ABSTRACT

The production of cement depletes natural resources, consumes high energy and emits huge amounts of green house gases. It accounts for almost 7% of the global carbon dioxide emissions, as the production of one ton of ordinary Portland cement releases approximately one ton of carbon dioxide. Due to the severe environmental pollution and health hazards associated with the cement and construction industries, they are under the strict scrutiny from the governments and environmentalists.

Rice husk is an agricultural waste, whose natural degradation is restricted due to the irregular abrasive surface and high siliceous composition. It is not appropriate to be used as a feed for animals due to the low nutritional values. If dumped as landfill, they can take a lot of area and become a major challenge to the environment. If they are disposed by burning, the ashes can spread to the surrounding areas, create pollution and destroy the natural beauty. One of the possible solutions for the disposal of rice husk is to convert them into rice husk ash and incorporate them into cement based materials. The partial inclusion of rice husk ash (RHA) for cement is found to be durable, environmental friendly and economically viable.

This paper presents an overview of some of the published results on the successful utilization of rice husk ash as a supplementary cementitious material and the properties of such concrete at fresh and hardened stages. Studies indicate that there is a promising future for the use of rice husk ash in normal, high strength and self compacting concrete as it shows high strength, low shrinkage and permeability, high resistance to carbonation, chloride, sulfate and acidic environments. The summary and discussions provided in this paper should provide new information and knowledge on the applications of greener and sustainable rice husk ash concrete.

1. Introduction

Sustainability in construction sector is one of the pressing needs of the developing world owing to the diminishing natural resources and increasing CO₂ emissions resulting from the huge production of cement concrete. The yearly concrete production is approximately 10 billion cubic meters. The most important and costliest constituent of concrete is cement, which is the binding material. As per the report of the United States Geological Survey, approximately 4180 million tons of cement was produced in 2014 globally. Cement manufacturing accounts for almost 7% of the global carbon dioxide emissions, as the production of one ton of cement (ordinary Portland cement) releases one ton of carbon dioxide. If a part of the cement in concrete can be replaced by any supplementary cementitious materials, it can reduce the consumption of ordinary Portland cement and thus reduces the emission of green house gases associated with the production of cement [35,70,75,79].

The research for sustainability has fuelled the search for alternate

materials that may replace some of the natural resources. Several industrial by-products such as fly ash, condensed silica fume, blast furnace slag, copper slag, steel scrap, stone wastes, tire ash, fibres etc., and some of the agricultural by products like palm oil shells, bagasse ash, corn cob, elephant grass ash, wood waste ash, coconut shell & fibres, rice husk ash, tobacco waste, etc have been found useful as additions or substitutions to cement and/or aggregates [42,52,53,73,74,78]. The utilization of some of these by-products provides several advantages such as improved strength and durability properties, reduction in construction cost by the saving of cement and natural aggregates, also environmental benefits like reduction in carbon dioxide emissions and the easy disposal of the polluting waste materials [1,37,39,69,76,77].

Rice (scientific name: *Oryza sativa*. L.) is one of the leading food crops in the world. It was forecasted that the global rice production in 2016 would be 745.5 million tonnes. More than 90% of the world's rice is produced and consumed by Asia on about 11% of the world's cultivated land. The hull or husk is the natural sheath surrounding the rice grains during their growth, constitutes to about 20–25% of the mass of

* Corresponding author.

E-mail address: chaprathu44@gmail.com.

Table 1

The world's forecasted production of rice paddy, potential husk and ash production for 2016 ([63]).

Country	Rice paddy production in Million Tons	Husk produced (20% of rice paddy) in Mt	Potential ash (18–20% of husk) in Mt
China	211.2	42.24	7.60
India	158.4	31.68	5.70
Indonesia	71.9	14.38	2.59
Bangladesh	52.9	10.58	1.90
Vietnam	44.5	8.90	1.60
Thailand	30.3	6.06	1.09
Africa	29.1	5.82	1.05
Myanmar	28.0	5.60	1.01
South America	23.7	4.74	0.85
Philippines	18.7	3.74	0.67
Japan	10.6	2.12	0.38
North America (US)	10.4	2.08	0.37
Pakistan	9.9	1.98	0.36
Cambodia	9.4	1.88	0.34
Europe	4.2	0.84	0.15
Others	32.3	6.46	1.16
Area (ha)	165 million hectares		
Total (world)	745.5 Million Tons		

the rough rice. The process of separating (shearing) the hull from the brown rice is known as dehulling. Rice milling is one of the most important industries in countries like India, China, Indonesia, Bangladesh and Vietnam. Table 1 shows the top 15 countries that have highest production of rice paddy. As per the forecast for 2016, China is expected to produce about 211.2 million tons of rice paddy every year, while India produces about 158.4 million tons of rice paddy every year. Out of the 158.4 million tons of rice paddy produced in India, almost 31.8 million tons will be rice husk and when burnt, it produces approximately 5.70 million tons of rice husk ash [62,8]. This can give an idea that each ton of paddy rice can produce approximately 200 kg of rice husk, which on combustion can produce about 40 kg of ash [9,45]. The rice producing countries are facing a severe problem on the disposal of rice husks. If they are dumped as landfill, they can take a lot of area and become a major challenge to the environment. If they are disposed by burning, the ashes will spread to the surrounding areas, create pollution and destroy the beauty of the land. When rice husk is fermented by micro organisms, it enables the emission of methane and thus contributing to the global warming problems [3,6,14,21,87].

It is not appropriate to use the rice husk as a feed for animals due to the low nutritional values. Moreover, natural degradation of rice husk is restricted due to the irregular abrasive surface and its high siliceous composition, which can make it a potential candidate for environmental pollution. In many countries, it has been widely and effectively used as a fuel for rice paddy milling process and power plants. This burning process creates a pozzolanic material named rice husk ash in which more than 75% by weight is silica (After the incineration process, 20% of the weight of the rice husk remains in the form of rice husk ash). The ash this generated has no immediate useful application and was usually dumped into water streams, which causes environmental pollution and river contamination. If the rice husk is converted to ash at a temperature below 500 °C, the ignition will not be complete and the pozzolanic effect of the ash will be minimized due to the presence of unburnt carbon. When the ash is produced at a controlled temperature of 550–700 °C, the silica content will be transferred to non-crystalline or amorphous silica (the silica content is approximately 90%) and is highly suitable for the use as a pozzolan [24,25,89]. Fig. 1 shows rice husk, unground and ground RHA.

According to Malhotra and Mehta [50], pozzolans are defined as siliceous or siliceous and aluminous materials which in themselves possess little or no cementing property, but will in a finely dispersed

form in the presence of water chemically react with calcium hydroxide at ordinary temperature to form compounds possessing cementitious properties. When compared to all other agricultural waste products, rice husk ash contains the higher proportion of silica. It was observed by Kamia et al. (2000) that the rice plants ingest orthosilicic acid from ground water and then it is polymerised to form amorphous silica on the husks. The RHA thus produced by means of controlled combustion, can react with the calcium hydroxide present in cement in the presence of water resulting in cementitious compounds [24,59,89].

Since the end of 1960s, a lot of research has been carried out on the utilization of RHA in cement concrete as a partial substitute to ordinary Portland cement and/or fine aggregates and many papers and patents have been published [54,55,62,66]. As per the available literatures, the research on RHA on cementitious materials was initiated by Mehta, who obtained a patent in 1973. His research work was on the effect of pyro-processing on the pozzolanic reactivity of RHA. Based on the work of Mehta, another research was carried out by Pitt (patent on 1976) who developed a fluidized bed furnace for the controlled combustion of rice husk. He identified that highly pozzolanic RHA could be produced when the burning temperature and the residency time of rice husk inside the furnace are controlled [60]. Since then, a lot of research on the application of RHA was being carried out throughout the world.

2. Properties of rice husk ash (RHA)

Rice husk ash is a good pozzolan and the color ranges from white grey to black, depending up on the source of raw material, method of incineration, time, duration and temperature of burning. The raw rice husk will be fired in a furnace with a controlled temperature starting at 10 °C per minute and increases gradually up to 700 °C per minute for about six hours and then allowed to cool at room temperature. It was observed by James and Rao [36] that isothermal heating at a minimum of 402 °C is required for the liberation of silica after the complete destruction of the organic matter from rice husk. The most reactive silica was obtained 500 °C combustion [36,59]. Bie et al. [7] studied on the effects of burning conditions (of RHA in a muffle furnace) on the mechanical behaviour of cement. They concluded that the rice husk burn completely at 600 °C than at high temperature. When the temperature was increased, K₂O could decompose and create more carbon in RHA. ASTM C618 states the requirements for pozzolan to be used in cement and concrete. As per the code, RHA is classified as type N pozzolan. The combined amount of silicon dioxide, iron oxide and aluminium oxide should not be less than 70% of the total mass and the loss of ignition can be up to 10% [3]. The chemical composition of rice husk ash is given in Table 2 and the physical properties are given in Table 3. Salas et al. [66] mentioned that the specific surface of 274,000 m²/kg (which is 10 times that of silica fume and normal RHA) can be achieved in RHA if it is treated. The rice husk was treated with hydrochloric acid at a concentration of 1 N for 24 h and then washed thoroughly with water to make neutral pH. The husk was then dried and burned to ash in a muffle furnace at a temperature of 600 °C for 3 h.

3. Properties of concrete containing RHA

3.1. Fresh concrete properties

3.1.1. Workability and bulk density

Fresh concrete is a plastic concrete that can be moulded to any shape. Hundred per cent compaction of fresh concrete is an important parameter to enable maximum strength for concrete. A highly workable concrete can ensure full compaction. Workability of concrete is the ease with which concrete can be mixed, handled and compacted. Bui et al. [10] studied the properties of gap graded aggregate concrete containing 10–20% RHA. It was observed that, at a low water to cement ratio of 0.30–0.34, the use of high binder contents along with RHA and super plasticizer has enabled to obtain a high workable, cohesive mix without

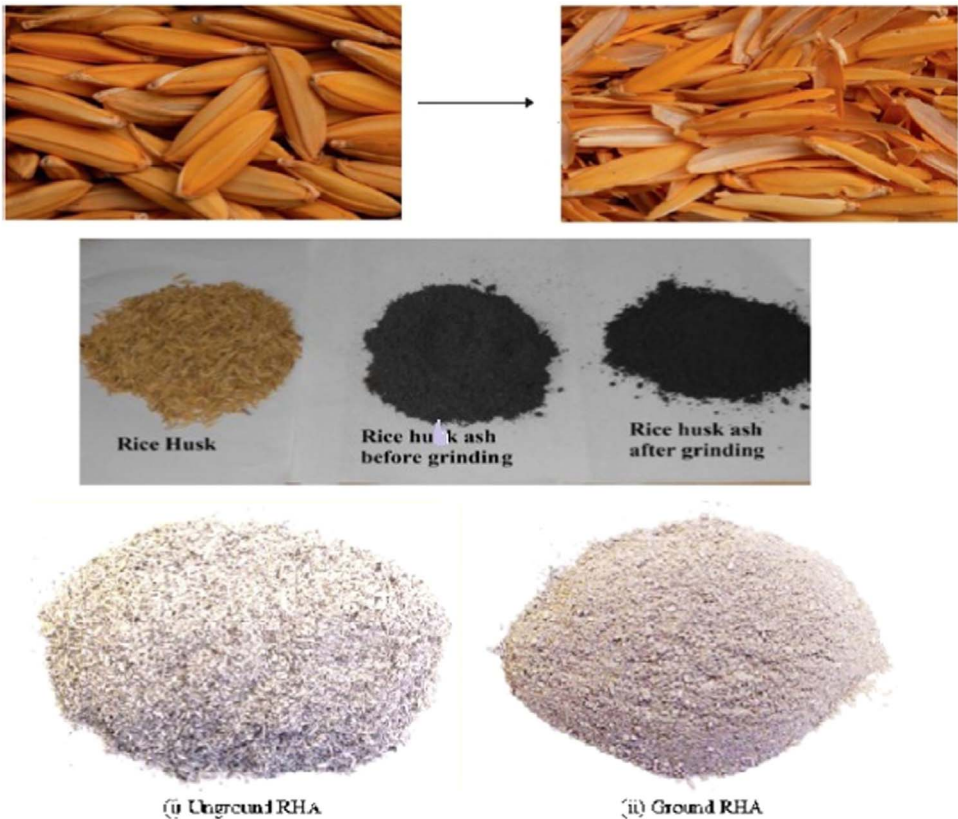


Fig. 1. Images showing rice husk, unground and ground RHA [13,84,8].

the problem of segregation. The increased water demand due to the addition of RHA was compensated by the use of superplasticizer. Similar observations were also made by Celik and Canakci [11] and Le et al. [46]. Salas et al. [66] determined the workability of rice husk ash concrete by slump method. They noticed that the concrete mixes are becoming stiffer at 15% and 20% RHA and hence it requires additional super plasticizer to maintain the required workability. Also those mixes showed the tendency of bleeding and segregation, which can affect the strength of the concrete. Safiuddin Md. and Soudki [65] observed an increase in the slump flow, inverted slump cone flow spread and orimet flow spread without and with j-ring at lower water-cement ratio and higher RHA content. The sign of segregation was observed in the RHA concrete, while the bleeding has reduced significantly. It was identified by Sua-iam and Makul [72] that the use of high volume RHA as a partial substitute for fine aggregate decreased the workability and segregation,

where the concrete containing RHA up to 25% maintained satisfactory workability. Antiohos et al. [2] reported that the target workability in RHA concrete can be achieved by the addition of a small amount of super plasticizer. Kannan and Ganesan [40] mentioned that the self compacting concrete containing 5–25% RHA fell under the category of slump flow classes 1 and 2, hence they may be suitable for deep foundation construction (SF1) and in normal applications like column construction (SF2). From the V-funnel test, the concrete containing 5–15% RHA could be categorized to VF1 class, while the entire mixes (from 5% to 30% RHA) fell within the ‘recommended’ category as the V-funnel time does not exceed 25 s. Satisfactory blocking ratio has been obtained for the SCC mixes containing 5–15% RHA, from the L-box test. So, the concrete mixes containing 5–15% RHA exhibited good workability and it can be implemented in self compacting concrete. It was mentioned by Le and Ludwig [47] that RHA can be used as a viscosity

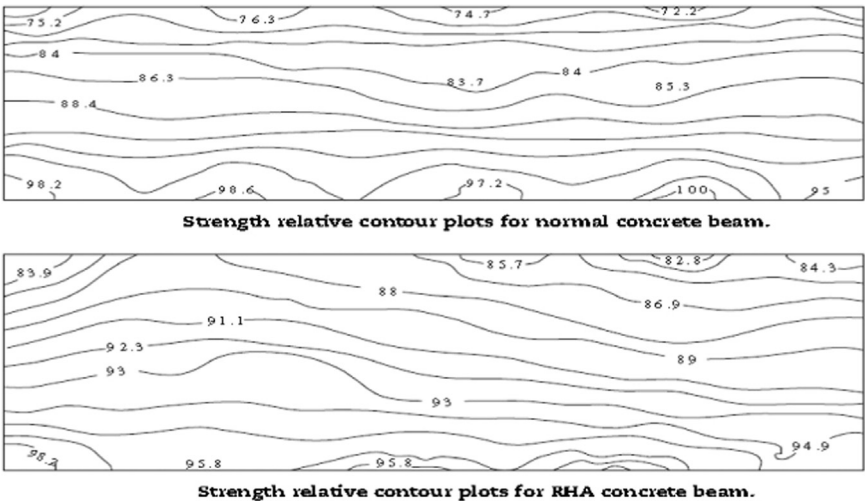


Fig. 2. Contour plots for strength variations for normal and RHA concrete by UPV method [48].

Table 2
Chemical composition of rice husk ash ([24,88,48,2,83,61,15,32]).

Chemical composition	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	MnO	Na ₂ O	K ₂ O	SO ₃	Loss on ignition
Value %	87.4–93.15	0.13–0.83	0.1–0.6	0.80–1.10	0.2–0.6	0–0.2	0.04–0.7	1.32–3.39	0.1–0.4	1.9–8.5

modifying admixture. The incorporation of RHA eliminated the bleeding tendency of cement mortar and that of self compacting high performance concrete.

Yuzer et al. [87] studied the concrete samples in which rice husk was added at 0%, 1.5%, 3% and 5% by weight of cement. It was observed that the addition of rice husk decreases the density of concrete. The density of control concrete was 2430 kg/m³. It reduced to 2340 kg/m³, 2260 kg/m³ and 2105 kg/m³ when the amount of rice husk was 1.5%, 3% and 5% respectively. Hamzeh et al. [34] investigated the density of cement concrete containing 25% and 50% white rice husk ash along with pulp fibres. There was a linear reduction in the bulk density of concrete with the increase in the ash content. It was stated by the authors that the addition of white RHA can be very much effective for reducing the density of light weight cement composites.

3.2. Hardened concrete properties

3.2.1. Compressive strength

Gastaldini et al. [28] mentioned that the compressive strength of the concrete containing 20% RHA with a water to cement ratio of 0.35 exhibited a compressive strength of 75.2 MPa at 28 days. The concrete containing 20% RHA and 1% K₂SO₄ (by weight of cement, as a chemical activator) have exhibited an increase in compressive strength (when compared to control concrete) by 9%, 33% and 43% for the mixture with water to cement ratios of 0.35, 0.50 and 0.65 respectively at the age of 91 days. Similar results were observed by Chindaprasirt et al. [16], who suggested that the optimum cement replacement with RHA can be 20%. Salas et al. [66] observed that the use of 5–10% RHA significantly increased the strength of the concrete and comparable with the concrete containing same amount of silica fume. Madandoust et al. [48] conducted studies by replacing 20% of cement with RHA and observed that the short term strength enhancement of RHA concrete is lower than the normal concrete. At the age of 3 days, the strength of RHA concrete was 65% of the normal control concrete. It increased to 96% at 90 days, 98% at 180 days, equivalent strength at 270 days and 102% (means 2% higher strength) at 360 days. Due to the higher pozzolanic activity in RHA concrete, it may be unacceptable in fast track construction projects owing to the reduced strength development at early age. Similar observations were made by Yuzer et al. [87]. Suaian and Makul [71] observed reduction in compressive strength when RHA was replaced for 0–100% of fine aggregates. The compressive strength reduced with increasing amount of substitution. Antiohos et al. [2] examined the compressive strength of mortar specimens by using RHA (0–30%) of Blaine fineness 4000 and 7000 cm²/g. It was observed that the samples containing RHA of 7000 cm²/g fineness outperformed the first one in terms of strength development. At replacement level of 10%, the strength of RHA concrete was similar to that of control concrete and the strength decreased upon further replacement. Kannan and Ganesan [40] recommended the combination on RHA and metakaolin for the production of self compacting concrete with higher cement replacement up to 40%. When 20% RHA and 20% metakaolin was used,

there was an increase of up to 30% in the compressive strength of SCC. It was pointed out by Bie et al. [7] that the use of 10% RHA can be the optimum proportion to enhance the strength of cement mortar. Zahedi et al., [88] mentioned that the use of RHA with nano silica would be beneficial as the nano silica can contribute towards the high early strength, while the RHA can increase the long term strength and durability. Kunchariyakun et al. [43] observed an increased water demand that caused a reduction in the compressive strength of the autoclaved aerated concrete, when RHA was used to partially replace fine aggregate. Siddique et al., [70] studied on the properties of RHA concrete containing bacteria (*Bacillus aerius* strain AKKR5). When compared to the control concrete, the compressive strength of 10% RHA concrete increased by 8.7%, 10% and 13.4% at 7, 28 and 56 days. The concrete containing RHA and bacterial cells had a compressive strength of 10.2%, 11.8% and 14.7% higher than the control concrete. This increase in strength can be due to the formation of calcite within the cement-sand pore structure.

3.2.2. Flexural and splitting tensile strength

It was observed by Salas et al. [66] that the flexural tensile strength (modulus of rupture) of the concrete specimens containing ordinary RHA are comparable to the control mix specimens, while the strength is much higher for the concrete containing treated RHA. Giaccio et al. [32] analysed the failure mechanism of RHA concrete and reported that there is a tendency for brittle failure as the stresses at which the cracks develop and propagate are higher in RHA concrete when compared with control concrete. Wei and Meyer [86] mentioned that the partial replacement by RHA improves the flexural properties and the durability to wetting and drying cycles of the fibre reinforced concrete composites. Similar observations were made by Chopra et al. [20]. Mohseni et al. [58] noticed a reduction in the 28-day flexural strength by 2%, 11% and 14% when RHA was replaced for 10%, 20% and 30% of cement. The flexural strength of the specimens containing 30% RHA increased at 90 days of age. When 3% nano-Al₂O₃ was added by weight of cement to the samples containing 10% RHA, there was an increase in the flexural strength by 34% and 41% at the ages of 28 days and 90 days respectively. When 0.3% polypropylene fibre was added to the concrete containing 10% RHA and 1% nano-Al₂O₃, the 28 day and 90 day flexural strength increased by 18.6% and 23.1% respectively.

Saraswathy and Song [69] carried out the splitting tensile strength as per ASTM C496-90 using cylindrical specimens of 150 mm diameter and 300 mm height. The strength was found to gradually increasing with the increasing amount of RHA, from 5% to 15% of substitution. The strength of control concrete was 4.49 N/mm² and it was 4.92 N/mm² for the specimens containing 15% RHA. When the substitution was 20–25%, the strength reduced to 4.60 N/mm² and 4.58 N/mm² respectively, still higher than the control mix values. The strength suffered a huge reduction at 30% substitution, where it reached to 3.67 N/mm². Madandoust et al. [48] observed that the splitting tensile strength of RHA concrete (20% replacement) was 79% of normal concrete at 3 days and it increased to 104% at 365 days. Madandoust and

Table 3
Physical Properties of rice husk ash ([24,88,48,2,83,61,15,32]).

Property	Specific gravity	Blaine specific surface, m ² /kg	Pozzolanic activity index (%)	Bulk density, Kg m ³	Median particle size (μm)	Nitrogen adsorption, m ² /kg	Surface area, cm ² /g
Value	2.06–2.16	240.0–376.8	81.25–88.90	420.0–429.1	5.0–7.41	24.3–28.8	4091–5685

Ghavidel [49] stated that the initial splitting tensile strength of control concrete was higher than that of the blended concrete containing 10% glass powder and 5% RHA by weight of cement. The strength of the blended concrete increased with age as it was 71% of the control concrete at the age of 7 days and it increased to 97% at the age of 90 days. Venkatanarayanan and Rangaraju [84] have observed an increase in strength comparing to control concrete; when the level of rice husk ash was 7.5% and 15%. It was 16% and 4% when unground rice husk ash was used, while the increase was 21% and 15% when ground RHA was used.

3.2.3. Abrasion resistance and fatigue life

The abrasion resistance of concrete may be defined as its ability to resist being worn away by rubbing. If there is reduction in the depth of abrasion, it means that the concrete is more resistant to abrasion. The concrete which is more resistant to abrasion can be applied in pavements, floors and concrete highways, in hydraulic structures such as tunnels and dam spillways, or in other surfaces upon which abrasive forces are applied between surfaces and moving objects during service [31]. Siddique et al. [70] determined the abrasion resistance of concrete as per BIS 1237 using concrete specimens of 70.6×70.6 mm size. The depth of abrasion in 10% RHA concrete was 0.831, 0.585 and 0.491 mm at 7, 28 and 56 days whereas the depth of wear was only 0.752, 0.501 and 0.400 mm at 7, 28 and 56 days in the case of concrete containing 10% RHA and bacteria.

Jamil et al. [38] identified the optimum replacement level of RHA as 14.3% for ASTM type-1 cement with 55% C_3S and 19% C_2S . Modarres and Hosseini [56] observed that the addition of 3% RHA reduced the porosity of concrete after 120 days of curing and thus improved the fatigue resistance and energy absorbency of the material, while the addition of 5% RHA resulted in higher porosity and lower fatigue lives of the concrete.

3.2.4. Water absorption

Controlling the permeability of concrete plays an important role, as the ingress and movement of ions from the environment, through the building materials are mainly responsible for the deterioration of structures. Saraswathy and Song [69] have noticed a reduction in porosity of concrete for up to 30% substitution of OPC with RHA. The effective porosity was 18.06% for control concrete; 13.80% for the specimens containing 15% RHA and 11.89% for the specimens containing 30% RHA. Salas et al. [66] performed the water absorption test of control and RHA concrete by the method of immersion (ASTM C642). It was observed that the water absorption of the concrete containing normal RHA was lower than the control concrete and that of the concrete containing treated RHA was much lower than the concrete containing normal RHA. Hamzeh et al. [34] have noticed reduction in water absorption of the specimens upon increase in the amount of white rice husk ash, while there was a sharp increase in the water absorption when the amount of ash was more than 25% by weight of cement. Kannan and Ganesan [40] noticed a decrease in the sorptivity with the increasing amount of RHA content up to 15% in SCC. When the RHA content was 20% and above, the sorptivity increased due to the formation of micro pores, which is attributable to the higher surface area of the RHA and the subsequent water demand. When compared to the sorptivity value of the control concrete specimens, there was a reduction of 19.10% in the sorptivity of concrete containing 20% RHA + 20% metakaolin.

Zahedi et al. [88] observed a 14.6% reduction in the water absorption of the mortar samples containing 20% RHA, while the reduction was 28.1% for the specimens containing a combination of nano silica and 10% RHA. Siddique et al. [70] have noticed water absorption of 2.51%, 1.66% and 1.47% at 7, 28 and 56 days respectively, in case of concrete containing 10% RHA. When the pores were filled by calcite produced by bacteria, it further decreased the water absorption of the RHA concrete to 1.86%, 1.03% and 0.80% at 7, 28 and 56 days.

Venkatanarayanan and Rangaraju [84] mentioned that the rate of water absorption of the concrete specimens containing 0–15% unground rice husk ash was almost constant, while there was a decrease in the water absorption upon increase in the amount of ground rice ash in concrete. When compared to the control specimens, the decrease in water absorption was 13% when the ground RHA was at 7%, and it was 12% for 15% ground RHA. Mohseni et al. [58] studied on the durability properties of cement mortars incorporating rice husk ash (0–15%) and nano- TiO_2 particles (1%, 3% and 5% of the binder). The water absorption reduced with the increasing amounts of RHA and NT. Due to the denser micro structure, there was about 30% reduction in the water absorption of the specimens containing 15% RHA and 5% NT.

3.2.5. Air permeability, modulus of elasticity and radon resistant potential

De Sensale [24] studied the air permeability of concrete (Swiss Standard SIA 262/1:2003) containing 0–15% RHA. Three different water to cement ratios (0.5, 0.4 and 0.32) were studied. At a water to cement ratio of 0.5, the concrete samples containing RHA were better than those without RHA and the quality of concrete was termed as 'good'. At water to cement ratios of 0.4 and 0.32, in most of the substitution levels, the quality of concrete with and without RHA is 'good', while the concrete mixes containing 10% and 15% RHA substitution was in 'very good' quality level.

Venkatanarayanan and Rangaraju [84] determined the 28 day modulus of elasticity of concrete as per ASTM C469 using concrete cylindrical specimens of size 100 mm $\times$ 200 mm. The modulus of elasticity of concrete specimens containing unground and ground rice husk ash was higher than that of the control specimens by 9–16%. It was observed that the grinding effect of RHA have a negligible impact on the modulus of elasticity of concrete. The addition of unground rice husk ash at 7.5% and 15% increased the modulus of elasticity by 11% and 13% respectively, while the addition of same amounts of ground rice husk increased the modulus of elasticity of concrete by 10% and 16% respectively.

Out of all radioactive sources, approximately 55% of the total dose was contributed by radon and the radon from the soil diffuse through the cracks and pores of the concrete by advective flow. The diffusion of radon through the concrete can be prevented to a great extent by the application of anti radon coating, choice of proper building materials etc [13,82]. It was observed by Chauhan and Kumar that the effective radon diffusion coefficient and diffusion length of the cement mortars decreased with increasing percentages of rice husk ash up to 30% of replacement. Beyond 30%, the radon diffusion increased and it may be due to the decrease in the density. The diffusion coefficient of the concrete containing 30% rice husk ash was reduced by 69% when compared to the control concrete specimens.

3.2.6. Chloride penetration

Gastaldini et al. [28] have observed a significant reduction in the total charge passed by 65%, 68% and 59% at water to cement ratios of 0.35, 0.50 and 0.65 respectively (at 20% RHA). When the concrete containing 20% RHA was chemically activated by 1% K_2SO_4 by weight of cement, there was a decrease in the total charges by 75%, 74% and 71%. When the concrete mixtures was activated by 1% Na_2SiO_3 , the total charges were reduced by 74%, 73% and 75% at water to cement ratios of 0.35, 0.50 and 0.65 respectively. Salas et al. [66] performed the rapid chloride permeability test (ASTM C1202) to measure the electrical conductivity of concrete. It was observed that the transferred charge was very low in RHA concrete, as the presence of rice husk ash helped to densify the pore structure of the concrete. The pore refinement was confirmed by mercury intrusion porosimetry (MIP) technique which confirmed the average diameter of 0.041 μm for control cement paste and 0.021–0.023 μm for the concrete containing RHA. Similar studies were reported by Chindaprasirt et al. [19] and Antiohos et al. [2]. Ganesan et al. [27] replaced 30% OPC with RHA and observed about 35% reduction in water permeability, 28% reduction in chloride

diffusion and 75% reduction in chloride permeation, when compared to the control concrete. De Sensale [24] suggested that the best resistance to chloride ion penetration (determined by silver nitrate spraying test) can be obtained by substitution of 15% of Portland cement by RHA. Madandoust et al. [48] tested the chloride ion penetration as per ASTM C1152 on the specimens containing 20% RHA. Samples were exposed to 5% NaCl with wet and dry cycling for a period of eleven months. Powder was taken from the samples after dry drilling (0–10 mm, 10–20 mm, 20–30 mm and 30–40 mm). From the chemical analysis, it was shown that there was lower chloride concentration with the higher amount of RHA in concrete and the penetration decreases with concrete depth. Siddique et al. [70] observed that the charge passed by the 10% RHA concrete decreased by 52.6% at 56 days of curing when compared to control concrete. In the case of bacterial concrete, it decreased by 48.4% at 56 days when compared to bacterial control concrete. The permeability range for RHA concrete was from 'low' to 'moderate' while it was from 'very low' to 'moderate' in case of bacterial concrete containing RHA.

3.2.7. Resistance to acid attack

Degradation can take place if the concrete is exposed to aggressive sulfuric acid environments. It is one of the key durability issues that affect the maintenance costs and life cycle performance of all the concrete structures. There can be presence of sulfuric acid in chemical waste, ground water, etc. In the case of concrete structures in industrial zones, there can be possibility of deterioration due to acid rains in which sulfuric acid can be one of the key components. Sulfuric acid attack is more disastrous than sulfate attack because of the fact that there would be a dissolution effect by the hydrogen ions in addition to the attack by sulfate ions [5]. De Sensale [24] evaluated the resistance of RHA concrete to acid attack as per ASTM C267. The cylindrical mortar specimens of 50×100 mm were immersed in a solution containing 1% HCl. The mass of specimens were determined (up to 84 days) at every 7 days of immersion, in saturated surface dry condition. It was observed that the percentage of mass loss of reference mortars increases rapidly when compared to the RHA concrete. As the percentage of cement substitution with RHA is increased, the acid attack decreases. When the amount of RHA was increased in the mortar, more calcium hydroxide (which is readily attacked by HCl) has been converted to C-S-H and thus increasing the resistance to acid attack.

Kannan and Ganesan [40] noticed that there was more loss in weight of the specimens immersed in 5% sulfuric acid than those immersed in hydrochloric acid (shown in Fig. 3). The self compacting concrete containing RHA + metakaolin exhibited better resistance to acid attack than the normal RHA SCC. Chatveera and Lertwattanakul [12] investigated the resistance of concrete (containing 20% and 40% black rice husk ash.) to HCl and H_2SO_4 . When the amount of rice husk ash was 20%, there was a positive effect in reducing the degradation of concrete under HCl and H_2SO_4 attack. When 40% of the OPC was replaced by RHA, the resistance to HCl attack improved, while the

resistance to H_2SO_4 was impaired and the specimens suffered more loss in weight. Similar observations were made by Khan et al. [41]. Chatveera and Lertwattanakul [12] studied the performance of black RHA mortar specimens in nitric and acetic acid with 5% concentration. It was observed that the use of RHA has a positive effect in reducing the expansion, loss in compressive strength and weight of mortar under acetic and nitric acid attacks. More loss was observed in the specimens exposed to acetic acid than under nitric acid and lowest loss in weight and strength was observed in the specimens containing 10% RHA. The rate of expansion of mortars decreased at 10–30% replacement of RHA, while it increased at 40–50% substitution. Increasing the water to cement ratio had a negative impact on expansion, compressive strength and weight of mortars exposed to acetic and nitric acids.

3.2.8. Resistance to sulfate attack

Sulfates reacts with the hydrate of calcium aluminates to form calcium sulphoaluminates (the volume will increase approximately 227% of the volume of the original aluminates) and with the free calcium hydroxides in cement to form calcium sulfate. This reaction leads to the formation of gypsum, ettringite and/or thaumasite and causes detrimental effects like spalling, cracking, softening, expansion loss of strength and other forms of concrete damage. The expansion results in the formation of cracks and subsequent disruption of concrete. This phenomenon is known as the sulfate attack. Chindaprasirt et al. [17] observed good resistance to sulfate attack when RHA was replaced up to 40% of OPC. De Sensale [24] performed the sulfate attack test as per ASTM C1012 on mortar specimens of $25 \times 25 \times 285$ mm size. After 28 days of curing in a solution containing saturated lime-water, the length of the specimens was measured. The control specimens were then immersed in saturated lime water and the test specimens were immersed in a solution of 5% sodium sulfate. The change in length of the specimens was measured in every week, up to 200 days (28 weeks). At 28 weeks, the specimens containing RHA was observed to be more resistant to sodium sulfate attack and resistance increased with the increasing percentage of RHA in mortar specimens. This increase in resistance can be due to the following reason. When the concrete/mortar specimens are subjected to sulfate attack, the sodium sulfate reacts with calcium hydroxide and transfers it to gypsum. The gypsum reacts with the aluminium compounds to produce ettringite, which leads to the expansion of cement matrix and further cracking of the specimens. The amount of Al_2O_3 is much higher in cement, while it is negligible in RHA. So, increasing the amount of RHA can improve the resistance of concrete to sulfate attack [24,67,68].

3.2.9. Shrinkage, expansion and alkali-silica reaction

It was reported by Habeeb and Fayyadh [33] that the drying shrinkage of concrete samples containing RHA was higher than that of the control concrete sample. Similar results were observed by Gastaldini et al. [30]. De Sensale [24] studied the alkali-silica reaction using $25 \times 25 \times 285$ mm mortar specimens, as per ASTM C1260. The samples (16 days after casting) were immersed in 1 N NaOH solution for a period of 14 days and the variation in length was measured. Reference specimens has undergone higher amount of expansion. The expansion of mortar samples containing 10% RHA exceeded the limit of 0.10% expansion, while the expansion of the samples containing 20% RHA was within the limit. Zerbino et al. [89] studied the expansion of RHA concrete by accelerated expansion test as per ASTM C1293 using prism of $75 \times 75 \times 300$ mm size. It was observed that the concrete containing natural rice husk ash has undergone drastic expansions, while the expansion of concrete containing ground rice husk ash was within the limit of 0.04%. The use of ground rice husk ash has lead to the inhibition of alkali silica reaction. The damage levels due to the alkali silica reactions depend on the type of cement when the natural rice husk ash was used, as some of the cements can accelerate the reaction.

Van et al. [83] investigated the long term autogenous shrinkage of

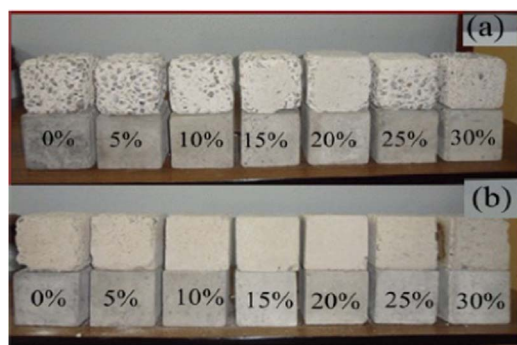


Fig. 3. RHA concrete specimens after 12 weeks of acid attack (a) in H_2SO_4 (b) in HCl [40].

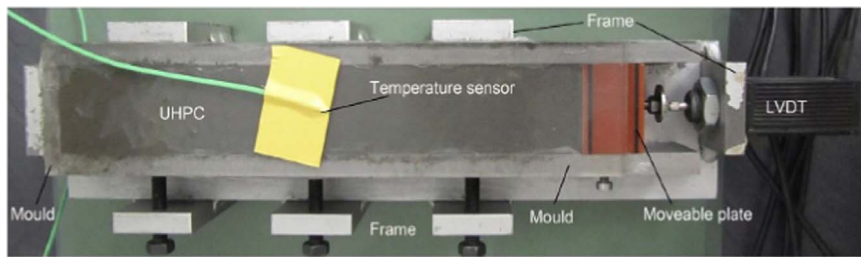


Fig. 4. Arrangement to measure the Autogenous shrinkage and temperature [83].

ultra high performance RHA concrete using three sealed $4 \times 4 \times 16 \text{ cm}^3$ sized specimen from two days of hydration, as per DIN 52450. The continual free length change and temperature was measured on the samples of $4 \times 4 \times 25 \text{ cm}^3$ size, up to 28 days of hydration as given in Fig. 4. The temperature of concrete and room temperature were measured by temperature sensors and the change in length of specimen was measured by linear variable differential transformers. It was observed that the addition of RHA significantly delays the self-desiccation of UHPC and thus prevents the autogenous shrinkage by behaving as an internal curing agent. When part of the cement was replaced by ground granulated blast furnace slag, the autogenous shrinkage of UHPC was found further reducing. Similar results were reported by Sensale et al. [23] who noticed that RHA markedly decreased the autogenous shrinkage as well as produces an autogenous relative humidity change. Chatveera and Lertwattanakul [12] have observed a negative effect on the drying shrinkage of concrete containing 20% black rice husk ash. Increasing the amount of rice husk ash from 20% to 40% by weight of the cement reduced the drying shrinkage and made it comparable to the control concrete specimens. The autogenous shrinkage of concrete with 20% RHA was lower than the control concrete. Increasing the rice husk ash from 20% to 40% further reduced the autogenous shrinkage of concrete. Mohseni et al. [58] have noticed reduction in the drying shrinkage of mortars containing RHA and poly propylene fibres, both separately and together. The drying shrinkage of control specimen was nearly 890 microstrains, while it was 729 micro strains for the specimen containing 3% poly propylene fibres, and 622 microstrains for the specimen containing 30% RHA and 3% polypropylene fibres.

3.2.10. Resistance to carbonation

Carbon dioxide can penetrate the surface of the concrete and react with the alkaline components, mainly $\text{Ca}(\text{OH})_2$ of the cement paste. This reaction (carbonation) can reduce the pH value of the pore solution to less than 9. When the alkalinity of the pore solution is lost, the protection of reinforcing steel against corrosion will be at risk. Also the properties of concrete like strength, permeability, shrinkage and resistance to chemical & physical attacks will be affected by carbonation [22]. Gastaldini et al. [28] examined the carbonation of chemically activated RHA concrete using $100 \times 100 \text{ mm}$ size specimens, as per RILEM TC 116-PCD and CPC 18, RILEM. The prepared specimens were exposed at 5% carbon dioxide inside a controlled chamber maintained at $23 \pm 1^\circ \text{C}$ and relative humidity $65 \pm 1\%$. The depth of carbon dioxide penetration was measured by means of phenolphthalein spraying test method (as shown in Fig. 5). It was observed that the depth of carbonation decreased when 20% RHA was used. The carbonation further decreased when 1% K_2SO_4 was used as a chemical activator along with 20% RHA. In the specimens with water to cement ratio of 0.5, the carbonation dropped from 7.50 to 4.69 mm and in the specimen with water to cement ratio of 0.65, it dropped from 14.14 to 10.68 mm. Chatveera and Lertwattanakul [12] investigated the depth of carbonation of concrete containing 20% and 40% black rice husk ash. The depth of carbonation of concrete containing 20% black rice husk ash was higher than that of the control concrete. Increasing the amount of ash from 20% to 40% and increase of water to cement ratio further increased the depth of carbonation. The increase in carbonation is attributed to the increase in porosity and the lower concentration of cement.

3.2.11. Electrical resistivity

Electrical resistivity is a method that is used for the quality control and performance based evaluation of concrete by measuring the diffusion of ions in the concrete through the pore solution. Layssi et al. [44] defined the electrical resistivity of concrete as the ability of the concrete to withstand the transfer of ions subjected to an electrical field. It is affected by the pore size distribution and interconnection, conductivity of pore fluid, degree of saturation and temperature. Electrical resistivity is measured by two-point uniaxial method and four-point (Wenner probe) method as shown in Fig. 6. Gastaldini et al. [29] have noticed an increase in the electrical resistivity as the curing period increased from 3 to 91 days. As the percentage of RHA in concrete was increased from 10% to 20%, the electrical conductivity doubled (in w/b ratios 0.35, 0.50 and 0.65) and at 30% RHA, the resistivity was 340%, 442% and 404% for the w/b ratios 0.35, 0.50 and 0.65 respectively when compared to the reference samples. Safiuddin Md. and Soudki [64] observed true electrical resistivity (between 5 and 10 $\text{k}\Omega \text{ cm}$) for all the concrete specimens that contain 0–20% RHA. The resistivity increased with increasing amount of RHA and at a content of 20%, it was 52% higher than the control concrete specimens. Chao-Lung et al. [15] noticed sharp increase in the electrical resistivity of the concrete containing ground RHA after 14 days of age (due to the solidification of the pores by pozzolanic reaction).

Zahedi et al. [88] studied the electrical resistivity of control mortar specimens, specimens containing RHA and the specimens containing both RHA and nano silica. The electrical resistivity of RHA concrete was found decreasing at the age of 7 days (due to the low early age pozzolanic activity) and the resistance increased at the age of 28 and 90 days when compared to the control mix. The highest electrical resistivity (improvement of 328.6% when compared to control mix) was observed in the specimens containing nano silica and 10% RHA at the ages of 28 and 90 days. Mohseni et al. [58] observed more chances of corrosion in control specimens (resistivity was between 5 and 10 $\text{k}\Omega \text{ cm}$) when compared to the specimens containing RHA and nano- TiO_2 particles. The electrical resistivity of the specimens containing 10% RHA + 5% NT, 15% RHA + 3% NT and 15% RHA + 5% NT were above 20 $\text{k}\Omega \text{ cm}$. As per the limits suggested by ACI Committee 222, the possibility of corrosion is low when the electrical resistivity is above 20 $\text{k}\Omega \text{ cm}$, low to moderate at 10–20 $\text{k}\Omega \text{ cm}$, high at 5–10 $\text{k}\Omega \text{ cm}$ and very high when it is less than 5 $\text{k}\Omega \text{ cm}$.

3.2.12. Corrosion performance

Saraswathy and Song [69] studied the corrosion behaviour of RHA concrete by impressed voltage test and open circuit potential measurements. In the impressed voltage test, concrete specimens were immersed in 5% NaCl solution. An external stainless steel served as cathode and the embedded steel in concrete was made anode. A constant positive potential of 12 V was supplied to the system from a DC source for up to 144 h and the time taken for initial crack along with the corresponding anodic current flow was measured. In the control concrete containing 100% OPC, cracks occurred after 42 h of exposure in the solution containing 5% NaCl. The concrete specimens containing 5% and 10% RHA failed at the exposure of 72 and 74 h respectively. No cracking was observed in the specimens containing 15%, 20%, 25% and 30% RHA, which may be due to the pore refinement by the finer RHA



Fig. 5. Samples showing the depth of carbonation [76,77,80].

particles. For the open circuit potential measurements (OCP), the specimens were subjected to alternate wetting and drying cycles (7 days of immersion in 3% NaCl solution and then 7 days of drying). As per ASTM C876, the embedded steel potentials were measured with reference to saturated calomel electrode (SCE). As per ASTM C876, if the potential values are lesser than -275 mV, the rebars are passive and vice versa. From the results it was observed that the rebars in all the specimens 0–30% RHA) were passive up to 100 days and beyond 100 days of exposure, all the rebars shifter to active condition irrespective of the replacement ratio of RHA.

Chindaprasirt and Rukzon [18] examined the corrosion behaviour of cement mortars of 40×40 mm dimension and 160 mm length) containing 0–40% RHA. The prisms were subjected to a 5% NaCl solution and a constant voltage of 12 V dc as given in Fig. 7. The time for first crack of control mortar was 89 h, while it was 167, 168 and 166 h for the mortars containing 10%, 20% and 40% RHA respectively. Ferraro and Nanni [26] studied the corrosion properties of concrete containing the environmental friendly off-white rice husk ash (contains no graphitic carbon, crystalline SiO_2 , and toxic metals). The first cracking of control samples started after 46 h while it was 74 h and 153 h in the concrete samples containing 7.5% and 15% off-white RHA respectively.

3.2.13. Thermal conductivity and water vapour diffusion resistance factor

Yuzer et al. [87] studied thermal conductivity and water vapour diffusion resistance factor of the concrete samples in which rice husk was added at 0%, 1.5%, 3% and 5% by weight of cement. The thermal conductivity of concrete was determined according to TS ISO 8302 on two specimens by using a guarded hot plate apparatus. It was observed that the thermal conductivity of concrete decreased with an increase in the amount of rice husk added. The thermal conductivity of control specimens was 1.88 W/mK, and decreased to 1.85 W/mK, 1.80 W/mK and 1.68 W/mK in concretes with 1.5%, 3.0% and 5.0% RH respectively. It can be because the rice husk present in the concrete produces empty pores that can reduce the heat transfer.

Water vapour diffusion resistance factor was determined as per TS

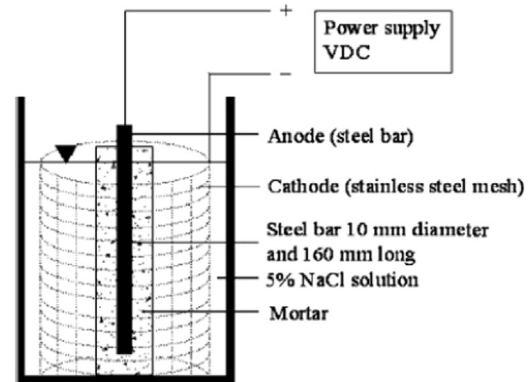


Fig. 7. Experimental set up for accelerated corrosion test with impressed voltage [18].

EN 12086. The specimens were kept in water-tight cups containing CaCl_2 as a sorbent. The water-tight cups were then placed in a controlled atmosphere cabinet at constant relative humidity and temperature. The specimens were weighed at an interval of 24 h to determine the quantity of moisture diffused through it. It was observed that the water vapour diffusion resistance factor decreased with an increasing amount of rice husk in concrete. The water vapour diffusion resistance factor for the control concrete specimens were 41, and it decreased to 32, 21 and 18 when the amount of rice husk was 1.5%, 3.0% and 5.0% respectively. This property can prevent the spalling of concrete containing rice husk, by reducing the vapour pressure within the concrete.

3.2.14. Micro structure by MIP, SEM and XRD

When RHA reacts with water and $\text{Ca}(\text{OH})_2$, it leads to the formation of calcium silicate hydrate (C-S-H) gel, which leads to the strength formation in all the cement based materials. Tuan et al. [81] measured the total porosity and porosity of RHA concrete using mercury intrusion

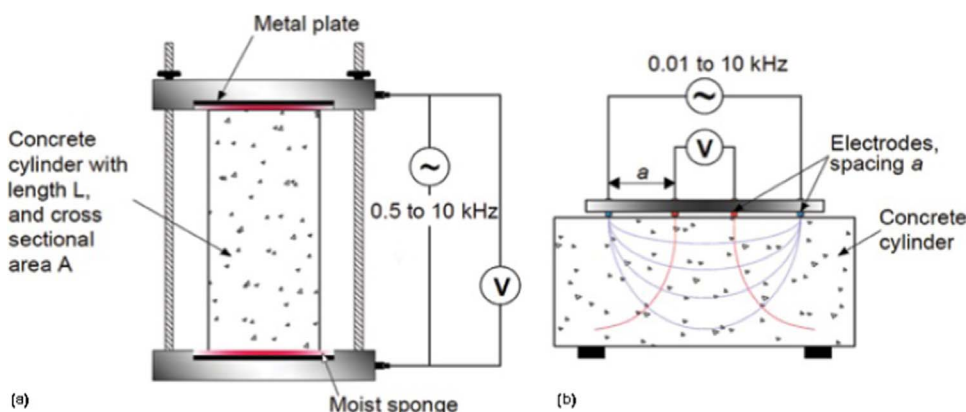


Fig. 6. Electrical resistivity measurement methods by (a) two-point uniaxial method (b) four-point (Wenner probe) method [44].

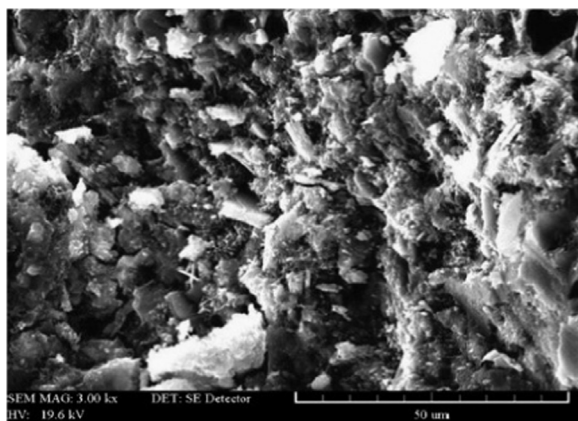


Fig. 8. SEM image of the RHA concrete showing flocc-like phases at 360 days [48].

porosimetry (MIP) technique. The critical pore diameter of the control sample was below $0.04\ \mu\text{m}$. Due to the high pozzolanic reaction and the filler effect of RHA, there was pore refinement and a reduction in the pore volume of the ultra high performance RHA concrete at the age of 28 and 91 days. It was also observed that the total porosity of RHA modified concrete was higher than that of the silica fume modified concrete as the pozzolanic reaction of silica fume has been higher than that of rice husk ash.

Madandoust et al. [48] analysed the microstructure of mortar specimens (by scanning electron microscopy) containing RHA, after 360 days of moist curing. As shown in Fig. 8, flocc-like hydration products are observed on the images. These hydration products were lapped and joined together by fibrous like formations. Kannan and Ganesan [40] noticed that the SEM image of the SCC without RHA consists of irregular particles with micro pores, while the concrete containing RHA was having more uniform structure (due to the reduction in the pores) and the ettringite was much reduced due to the pozzolanic reaction of the RHA. It was observed by Chopra et al. [20] that the particle shape of OPC is irregular and that of RHA is micro porous and irregular as shown in Fig. 9.

Siddique et al. [70] performed the X-ray diffraction of concrete samples containing 10% RHA (given in Fig. 10). The presence of calcium silicate hydrates (C-S-H), ettringite, calcium silicates, calcite and portlandite was shown in the XRD. Presence of fine aggregate (natural sand) was shown by the peaks of quartz (Q) and cristobalite (Cb). The presence of ettringite was noticed at 2 theta angles 35.1347° , 23.1374° , 15.9644° and 9.0185° . The typical XRD of RHA is given in Fig. 11.

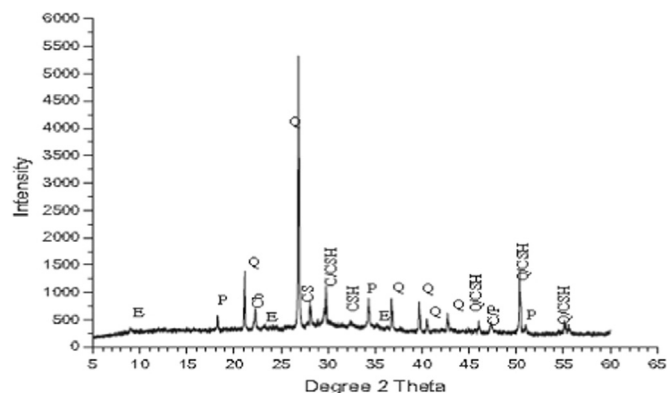


Fig. 10. X-ray diffraction of concrete containing 10% RHA [70].

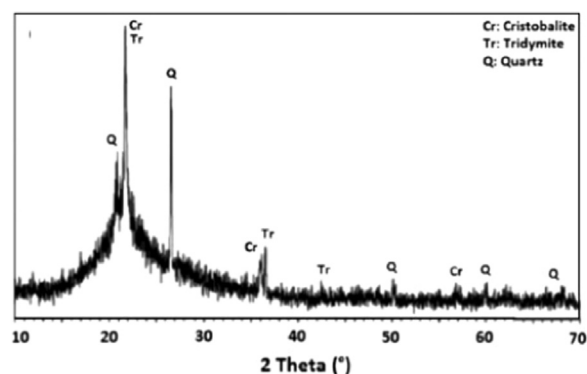


Fig. 11. XRD of rice husk ash [57].

Vishwakarma et al. [85] observed the peaks of silicon dioxide at 20.9° , 26.6° , 39.4° , 68.2° and 75.6° (JCPDS: 46-1045). The presence of calcium hydroxide peak was confirmed at 34.1° (JCPDS: 44-1481), calcium silicate hydrate at 50.1° (JCPDS: 33-0305) and the peak of calcium aluminium silicate hydrate at 27.9° (JCPDS: 39-1381).

3.2.15. Non destructive testing

Ultrasonic pulse velocity testing (UPV) have been used as a measure to assess the uniformity and relative quality of the concrete. The velocity of ultrasonic pulses travelling through a solid depends on the elastic properties and density of that material. It also helps indicate the presence of voids and cracks in concrete and to evaluate the effectiveness of crack repairs [4]. As per Malhotra [51], if the pulse velocity

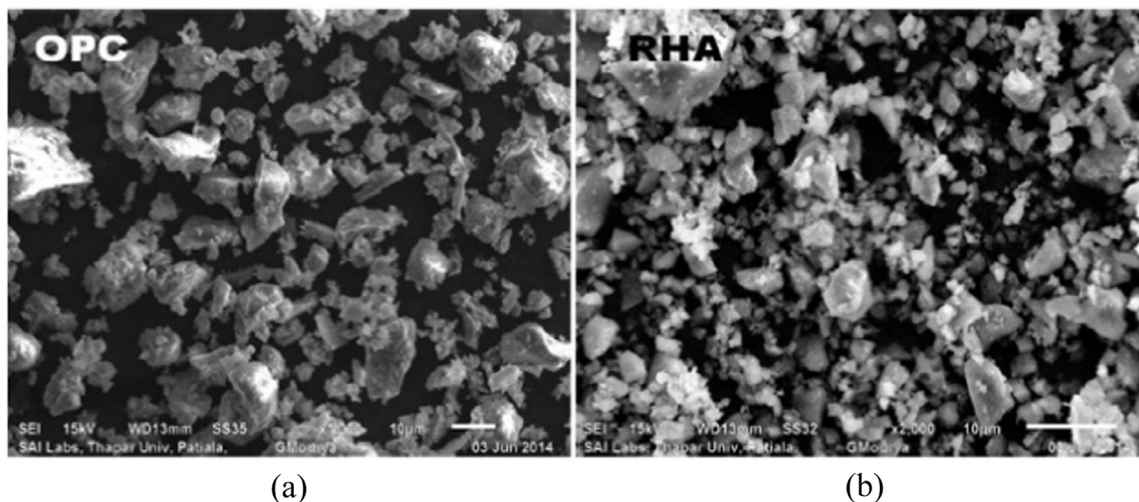


Fig. 9. SEM images of (a) ordinary Portland cement (b) Rice husk ash [20].

values are in the range of 3660–4575 m/s, it can be assumed that the concrete is having sufficient durability properties. Madandoust et al. [48] used ultrasonic pulse velocity of 54 kHz to assess the homogeneity of RHA and normal reinforced concrete beams. Least variation in strength development (approximately 13%) between the top and bottom layers was observed in RHA concrete, whereas the difference was about 23% in normal concrete. The contour line is shown in Fig. 2. It indicates that the concrete bonding can be improved by the addition of RHA and it further minimizes the effect of bleeding and segregation. Chao-Lung et al. [15] observed higher UPV values at lower water to binder ratios. The UPV of control concrete specimens and the specimens containing 20% RHA was the same. After 28 days of age, the UPV of the specimens containing 10–30% RHA was higher than 3660 m/s and hence they can be considered as durable concrete specimens. Sua-iam and Makul [71] examined the UPV of self compacting concrete containing 0–100% RHA as a partial to full substitute for fine aggregates. Highest velocity was observed in control concrete and the lowest velocity was observed in the samples containing 100% RHA. Mohseni et al. [58] suggested that the concrete containing 15% RHA and 5% nano-TiO₂ particles by weight of the binder can be the optimum as the pulse velocity value was above 4500 m/s.

4. Discussions and conclusions

1. The quality of rice husk ash depends on the source of raw material, method of incineration, time, duration and temperature of burning. Rice husk should be burned at a controlled temperature of 500–700 °C to produce the most reactive silica. Very high specific surface of 274,000 m²/Kg (10 times that of silica fume and normal RHA) can be achieved in RHA if it is treated with 1N Hydrochloric acid.
2. Rice husk ash can be used as a supplementary cementitious material with an amount of 5–10% substitute without compromising the workability. Beyond 10%, the workability reduces as the RHA demands more water, and this can be corrected by increasing the dosage of super plasticizer. The addition of RHA reduces the bleeding and segregation of mortars and cement concrete. The addition of RHA can be very much effective for reducing the density of cement composites.
3. Compressive strength of concrete increased with increasing amount of RHA, up to 20%. The concrete containing 20% RHA and 1% K₂SO₄ (by weight of cement, as a chemical activator) have exhibited an increase in compressive strength (when compared to control concrete). The use of RHA with nano-silica would be beneficial as the nano silica can contribute towards the high early strength, while the RHA can increase the long term strength and durability.
4. The resistance to abrasion, flexural and splitting tensile strength of concrete was found gradually increasing with the increasing amount of RHA, from 5% to 15% of substitution. There can be a tendency for brittle failure as the stresses at which the cracks develop and propagate are higher in RHA concrete when compared with control concrete. When the ultrasonic pulse velocity is considered, the concrete containing 15% RHA and 5% nano-TiO₂ particles by weight of the binder can be the optimum as researchers have obtained the pulse velocity value above 4500 m/s.
5. Concrete containing RHA up to 30% can exhibit a reduction in porosity and hence water absorption. Due to the denser micro structure, there can be about 30% reduction in the water absorption of the specimens containing 15% RHA and 5% nano-TiO₂ particles. The water absorption of concrete containing normal RHA can be lower than the control concrete and that of the concrete containing treated RHA (treated with HCl) could be much lower than the concrete containing normal RHA.
6. The concrete mixes containing 10% and 15% RHA substitution was in 'very good' air permeability quality level. The radon diffusion coefficient of the concrete containing 30% rice husk ash was

reduced by 69% when compared to the control concrete specimens. The modulus of elasticity of concrete specimens containing un-ground and ground rice husk ash was higher than that of the control specimens.

7. Increasing the amount of RHA in concrete by 15–20% reduces the corrosion potential, carbonation, chloride penetration, acid and sulfate attack. Increasing the amount of rice husk ash from 20% to 40% by weight of the cement reduced the drying shrinkage autogenous shrinkage of concrete and made it comparable to the control concrete specimens. The autogenous shrinkage of concrete containing 20% RHA could be lower than the control concrete.
8. The electrical resistivity of concrete containing RHA up to 30% would be much higher than the control concrete specimens. The resistivity at 7 days can be less, while a sharp increase in the electrical resistivity of the RHA concrete could be noticed after 14 days of age (due to the solidification of the pores by pozzolanic reaction). Researchers have noticed the highest electrical resistivity (improvement of 328.6% when compared to control mix) in the specimens containing nano silica and 10% RHA.
9. Due to the high pozzolanic reaction and the filler effect of RHA, pore refinement takes place and a reduction could be noticed in the pore volume of the RHA concrete at the age of 28 and 91 days. The SEM analysis of the RHA concrete shows more uniform structure (due to the reduction in the pores) and reduction in ettringite due to the pozzolanic reaction of the RHA.

References

- [1] Al-Khalaf MN, Yousif HA. Use of rice husk ash in concrete. *Int J Cem Compos Lightweight Concr* 1984;6:241–8.
- [2] Antiohos SK, Papadakis VG, Tsimas S. Rice husk ash (RHA) effectiveness in cement and concrete as a function of reactive silica and fineness. *Cem Concr Res* 2014;61–62:20–7.
- [3] Aprianti ES. A huge number of artificial waste material can be supplementary cementitious material (SCM) for concrete production e a review part II. *J Clean Prod* 2016. <http://dx.doi.org/10.1016/j.jclepro.2015.12.115>.
- [4] ASTM-C597. Standard test method for pulse velocity through concrete. *Annual Book of ASTM Standard* 04; 2002.
- [5] Bassuoni MT, Nehdi ML. Resistance of self-consolidating concrete to sulphuric acid attack with consecutive pH reduction. *Cem Concr Res* 2007;37:1070–84.
- [6] Bhattacharya SC, Joe MA, Kandhekar Z, Salam PA, Shrestha RM. Greenhouse-gas emission mitigation from the use of agricultural residues: the case of ricehusk. *Energy* 1999;43–59.
- [7] Bie R-S, Song X-F, Liu Q-Q, Ji X-Y, Chen P. Studies on effects of burning conditions and rice husk ash (RHA) blending amount on the mechanical behavior of cement. *Cem Concr Compos* 2015;55:162–8.
- [8] Buggenhout J, Brijis K, Celus I, Delcour JA. The breakage susceptibility of raw and parboiled rice: a review. *J Food Eng* 2003;117:304–15.
- [9] Bui DD. Rice Husk Ash as a mineral admixture for high performance concrete [Ph.D. thesis]. The Netherlands: Delft University of Technology; 2001.
- [10] Bui DD, Hu J, Stroeven P. Particle size effect on the strength of rice husk ash blended gap-graded Portland cement concrete. *Cem Concr Compos* 2005;27:357–66.
- [11] Celik F, Canakci H. An investigation of rheological properties of cement-based grout mixed with rice husk ash (RHA). *Constr Build Mater* 2015;91:187–94.
- [12] Chatveera B, Lertwattanaruk P. Evaluation of nitric and acetic acid resistance of cement mortars containing high-volume black rice husk ash. *J Environ Manag* 2014;133:365–73.
- [13] Chauhan RP, Kumar A. Radon resistant potential of concrete manufactured using Ordinary Portland Cement blended with rice husk ash. *Atmos Environ* 2013;81:413–20.
- [14] Chabannes M, Bénézet JC, Clerc L, Garcia-Diaz E. Use of raw rice husk as natural aggregate in a lightweight insulating concrete: an innovative application. *Constr Build Mater* 2014;70:428–38.
- [15] Chao-Lung H, Anh-Tuan BL, Chun-Tsun C. Effect of rice husk ash on the strength and durability characteristics of concrete. *Constr Build Mater* 2011;25:3768–72.
- [16] Chindaprasirt P, Homwuttiwong S, Jaturapitakkul C. Strength and water permeability of concrete containing palm oil fuel ash and rice husk-bark ash. *Constr Build Mater* 2007;21:1492–9.
- [17] Chindaprasirt P, Kanchanda P, Sathonsaowaphak A, Cao HT. Sulfate resistance of blended cements containing fly ash and rice husk ash. *Constr Build Mater* 2007;21:1356–61.
- [18] Chindaprasirt P, Rukzon S. Strength, porosity and corrosion resistance of ternary blend Portland cement, rice husk ash and fly ash mortar. *Constr Build Mater* 2008;22:1601–6.
- [19] Chindaprasirt P, Rukzon S, Sirivivatnanon V. Resistance to chloride penetration of blended Portland cement mortar containing palm oil fuel ash, rice husk ash and fly

- ash. *Constr Build Mater* 2008;22:932–8.
- [20] Chopra D, Siddique R, Kunal. Strength, permeability and microstructure of self-compacting concrete containing rice husk ash. *Biosyst Eng* 2015;130:72–80.
- [21] Chungsangunsit T, Gheewala SH, Patumsawad S. Environmental assessment of electricity production from rice husk: a case study in Thailand. *Electr Supply Ind Transit: Issues Prospect Asia* 2004.
- [22] CPC-18. Measurement of hardened concrete carbonation depth. RILEM Publications SARL; 1988.
- [23] De Sensale GR, Ribeiro AB, Gonçalves A. Effects of RHA on autogenous shrinkage of Portland cement pastes. *Cem Concr Compos* 2008;30:892–7.
- [24] De Sensale GR. Effect of rice-husk ash on durability of cementitious materials. *Cem Concr Compos* 2010;32:718–25.
- [25] De Sensale GR. Strength development of concrete with rice-husk ash. *Cem Concr Compos* 2006;28:158–60.
- [26] Ferraro RM, Nanni A. Effect of off-white rice husk ash on strength, porosity, conductivity and corrosion resistance of white concrete. *Constr Build Mater* 2012;31:220–5.
- [27] Ganesan K, Rajagopal K, Thangavel K. Rice husk ash blended cement: assessment of optimal level of replacement for strength and permeability properties of concrete. *Constr Build Mater* 2008;22:1675–83.
- [28] Gastaldini ALG, Isaia GC, Gomes NS, Sperb JEK. Chloride penetration and carbonation in concrete with rice husk ash and chemical activators. *Cem Concr Compos* 2007;29:176–80.
- [29] Gastaldini ALG, Isaia GC, Hoppe TF, Missau F, Saciloto AP. Influence of the use of rice husk ash on the electrical resistivity of concrete: a technical and economic feasibility study. *Constr Build Mater* 2009;23:3411–9.
- [30] Gastaldini ALG, da Silva MP, Zamberlan FB, Mostardeiro Neto CZ. Total shrinkage, chloride penetration, and compressive strength of concretes that contain clear-colored rice husk ash. *Constr Build Mater* 2014;54:369–77.
- [31] Gencel O, Sabri Gok M, Brostow W. Effect of metallic aggregate and cement content on abrasion resistance behaviour of concrete. *Mater Res Innov* 2011;15:116–23.
- [32] Giaccio G, de Sensale GR, Zerbino R. Failure mechanism of normal and high-strength concrete with rice-husk ash. *Cem Concr Compos* 2007;29:566–74.
- [33] Habeeb GA, Fayyadh MM. Rice husk ash concrete: the effect of RHA average particle size on mechanical properties and drying shrinkage. *Aust J Basic Appl Sci* 2009;3:1616–22.
- [34] Hamzeh Y, Ziabari KP, Torkaman J, Ashori A, Jafari M. Study on the effects of white rice husk ash and fibrous materials additions on some properties of fiber-cement composites. *J Environ Manag* 2013;117:263–7.
- [35] H.G. Oss, Minerals information – cement from United States Geological Survey; 2015. Online at: <<http://minerals.usgs.gov/minerals/pubs/commodity/cement/mcs-2015-cement.pdf>>.
- [36] James J, Rao MS. Silica from rice husk through thermal-decomposition. *Thermochim Acta* 1986;97:329–36.
- [37] James J, Rao MS. Reactivity of rice husk ash. *Cem Concr Res* 1986;16:296–302.
- [38] Jamil M, Kaish ABMA, Raman SN, Zain MFM. Pozzolanic contribution of rice husk ash in cementitious system. *Constr Build Mater* 2013;47:588–93.
- [39] Kamiya K, Oka A, Nasu H, Hashimoto T. Comparative study of structure of silica gels from different sources. *J Sol-Gel Sci Technol* 2000;19:495–9.
- [40] Kannan V, Ganesan K. Chloride and chemical resistance of self compacting concrete containing rice husk ash and metakaolin. *Constr Build Mater* 2014;51:225–34.
- [41] Khan R, Jabbar A, Ahmad I, Khan W, Khan AN, Mirza J. Reduction in environmental problems using rice-husk ash in concrete. *Constr Build Mater* 2012;30:360–5.
- [42] Kumar S, Gupta RC, Shrivastava S, Csetenyi L, Thomas BS. Preliminary study on the use of quartz sandstone as a partial replacement of coarse aggregate in concrete based on clay content, morphology and compressive strength of combined gradation. *Constr Build Mater* 2016;107:103–8.
- [43] Kunchariyakun K, Asavapisit S, Sombatsompom K. Properties of autoclaved aerated concrete incorporating rice husk ash as partial replacement for fine aggregate. *Cem Concr Compos* 2015;55:11–6.
- [44] Layssi H, Ghods P, Alizadeh AR, Salehi M. Electrical resistivity of concrete. *Concr Int* 2015;37:41–6.
- [45] Le HT, Siewert K, Ludwig H-M. Alkali silica reaction in mortar formulated from self-compacting high performance concrete containing rice husk ash. *Constr Build Mater* 2015;88:10–9.
- [46] Le HT, Kraus M, Siewert K, Ludwig H-M. Effect of macro-mesoporous rice husk ash on rheological properties of mortar formulated from self-compacting high performance concrete. *Constr Build Mater* 2015;80:225–35.
- [47] Le HT, Ludwig H-M. Effect of rice husk ash and other mineral admixtures on properties of self-compacting high performance concrete. *Mater Des* 2016;89:156–66.
- [48] Madandoust R, Ranjbar MM, Moghadam HA, Mousavi SY. Mechanical properties and durability assessment of rice husk ash concrete. *Biosyst Eng* 2011;110:144–52.
- [49] Madandoust R, Ghavidel R. Mechanical properties of concrete containing waste glass powder and rice husk ash. *Biosyst Eng* 2013;116:113–9.
- [50] Malhotra VM, Mehta PK. Pozzolanic and cementitious materials. Amsterdam: Gordon & Breach Publishers; 1996.
- [51] Malhotra VM. Testing hardened concrete: nondestructive methods. *Am Concr Inst Monogr Ser* 1976.
- [52] Mehra P, Gupta RC, Thomas BS. Properties of concrete containing jarosite as a partial substitute for fine aggregates. *J Clean Prod* 2016;120:241–8.
- [53] Mehra P, Gupta RC, Thomas BS. Assessment of durability characteristics of cement concrete containing jarosite. *J Clean Prod* 2016;119:59–65.
- [54] Mehta P.K. Siliceous ashes and hydraulic cements prepared therefrom, Belgium Patent 802909 (1973), U.S. Patent 4105459; 1978.
- [55] Mehta P.K. The chemistry and technology of cement made from rice husk ash. In: *Proceedings, workshop on rice husk ash cement, regional center for technology transfer, Bangalore, India; 1979*, p. 113–22.
- [56] Modarres A, Hosseini Z. Mechanical properties of roller compacted concrete containing rice husk ash with original and recycled asphalt pavement material. *Mater Des* 2014;64:227–36.
- [57] Mohseni E, Naseri F, Amjadi R, Khotbehsara MM, Ranjbar MM. Microstructure and durability properties of cement mortars containing nano-TiO₂ and rice husk ash. *Constr Build Mater* 2016;114:656–64.
- [58] Mohseni E, Khotbehsara MM, Naseri F, Monazami M, Sarker P. Polypropylene fiber reinforced cement mortars containing rice husk ash and nano-alumina. *Constr Build Mater* 2016;111:429–39.
- [59] Nair DG, Fraaij A, Klaassen AAK, Kentgens APM. A structural investigation relating to the pozzolanic activity of rice husk ashes. *Cem Concr Res* 2008;38:861–9.
- [60] Nehdi M, Duquette J, El Damatty A. Performance of rice husk ash produced using a new technology as a mineral admixture in concrete. *Cem Concr Res* 2003;33:1203–10.
- [61] Park KB, Kwon SJ, Wang XY. Analysis of the effects of rice husk ash on the hydration of cementitious materials. *Constr Build Mater* 2016;105:196–205.
- [62] Pitt N. Process for preparation of siliceous ashes, U.S. Patent 3959007; (1976).
- [63] Rice Market Monitor, 2016, Issue no 1. Annual report, Volume XIX, April 2016.
- [64] Safiuddin MD, West JS, Soudki KA. Hardened properties of self-consolidating high performance concrete including rice husk ash. *Cem Concr Compos* 2010;32:708–17.
- [65] Safiuddin MD, West JS, Soudki KA. Properties of freshly mixed self-consolidating concretes incorporating rice husk ash as a supplementary cementing material. *Constr Build Mater* 2012;30:833–42.
- [66] Salas A, Delvasto S, de Gutierrez RM, Lange D. Comparison of two processes for treating rice husk ash for use in high performance concrete. *Cem Concr Res* 2009;39:773–8.
- [67] Santhanam M, Cohen MD, Olek J. Mechanism of sulfate attack: a fresh look –part 1: summary of experimental results. *Cem Concr Res* 2002;32:915–21.
- [68] Santhanam M, Cohen MD, Olek J. Mechanism of sulfate attack: a fresh look –part 2: proposed mechanisms. *Cem Concr Res* 2003;33:341–6.
- [69] Saraswathy V, Song H-W. Corrosion performance of rice husk ash blended concrete. *Constr Build Mater* 2007;21:1779–84.
- [70] Siddique R, Singh K, Kunal, Singh M, Corinaldesi V, Rajor A. Properties of bacterial rice husk ash concrete. *Constr Build Mater* 2016;121:112–9.
- [71] Sua-iam G, Makul N. Utilization of limestone powder to improve the properties of self-compacting concrete incorporating high volumes of untreated rice husk ash as fine aggregate. *Constr Build Mater* 2013;38:455–64.
- [72] Sua-iam G, Makul N. Utilization of high volumes of unprocessed lignite-coal fly ash and rice husk ash in self-consolidating concrete. *J Clean Prod* 2014;78:184–94.
- [73] Thomas BS, Damare A, Gupta RC. Strength and durability characteristics of copper tailing concrete. *Constr Build Mater* 2013;48:894–900.
- [74] Thomas BS, Gupta RC, Kalla P, Csetenyi L. Strength, abrasion and permeation characteristics of cement concrete containing discarded rubber fine aggregates. *Constr Build Mater* 2014;59:204–12.
- [75] Thomas BS, Gupta RC. Long term behaviour of cement concrete containing discarded tire rubber. *J Clean Prod* 2015;102:78–87.
- [76] Thomas BS, Gupta RC, Mehra P, Kumar S. Performance of high strength rubberized concrete in aggressive environment. *Constr Build Mater* 2015;83:320–6.
- [77] Thomas BS, Gupta RC, Panicker VJ. Experimental and modelling studies on high strength concrete containing waste tire rubber. *Sustain Cities Soc* 2015;19:68–73.
- [78] Thomas BS, Gupta RC, Panicker VJ. Recycling of waste tire rubber as aggregate in concrete: durability-related performance. *J Clean Prod* 2016;112:504–13.
- [79] Thomas BS, Gupta RC. Properties of high strength concrete containing scrap tire rubber. *J Clean Prod* 2016;113:86–92.
- [80] Thomas BS, Gupta RC. A comprehensive review on the applications of waste tire rubber in cement concrete. *Renew Sustain Energy Rev* 2016;54:1323–33.
- [81] Tuan NV, Ye G, Breugel KV, Copuroglu O. Hydration and microstructure of ultra high performance concrete incorporating rice husk ash. *Cem Concr Res* 2011;41:1104–11.
- [82] UNSCEAR. United Nations Scientific Committee on the Effects of Atomic Radiation. Effects and Risks of Ionizing Radiations. New York: United Nations; 2000.
- [83] Van V-T-A, Rößler C, Bui D-D, Ludwig H-M. Rice husk ash as both pozzolanic admixture and internal curing agent in ultra-high performance concrete. *Cem Concr Compos* 2014;53:270–8.
- [84] Venkatanarayanan HK, Rangaraju PR. Effect of grinding of low-carbon rice husk ash on the microstructure and performance properties of blended cement concrete. *Cem Concr Compos* 2015;55:348–63.
- [85] Vishwakarma V, Ramachandran D, Anbarasan N, Rabel AM. Studies of rice husk ash nanoparticles on the mechanical and microstructural properties of the concrete. *Mater Today: Proc* 2016;3:1999–2007.
- [86] Wei J, Meyer C. Utilization of rice husk ash in green natural fiber-reinforced cement composites: mitigating degradation of sisal fiber. *Cem Concr Res* 2016;81:94–111.
- [87] Yuzer N, Cinar Z, Aköz F, Biricik H, Gurkan YY, Kabay N, Kizilkanat AB. Influence of raw rice husk addition on structure and properties of Concrete. *Constr Build Mater* 2013;44:54–62.
- [88] Zahedi M, Ramezaniapour AA, Ramezaniapour AM. Evaluation of the mechanical properties and durability of cement mortars containing nanosilica and rice husk ash under chloride ion penetration. *Constr Build Mater* 2015;78:354–61.
- [89] Zerbino R, Giaccio G, Isaia GC. Concrete incorporating rice-husk ash without processing. *Constr Build Mater* 2011;25:371–8.